HandPad: A Bimanual Hand Interface for Fluid Window Interactions in VR

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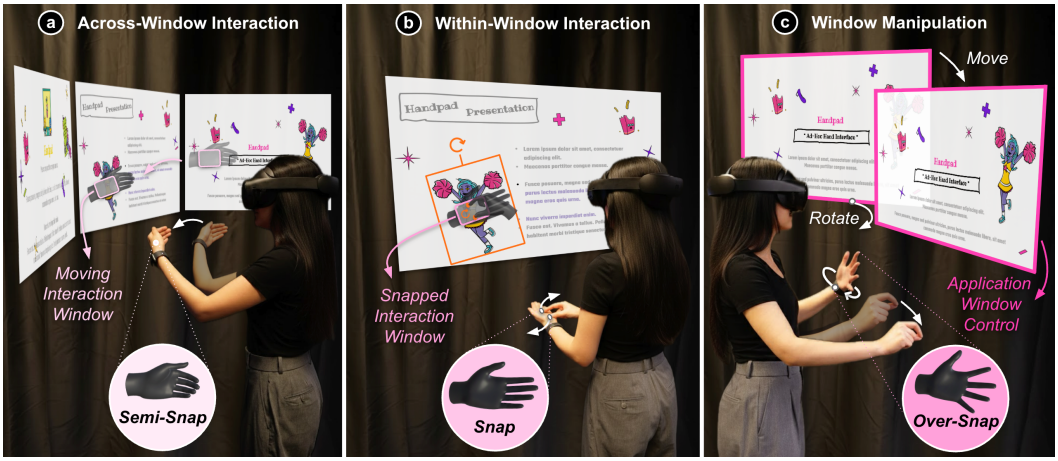


Fig. 1. HandPad utilizes the non-dominant hand (NDH) to establish spatial frames and interaction contexts, enabling the dominant hand (DH) to perform fine-grained manipulations. (a) Semi-Snap: A slightly bent NDH gesture initiates an interaction window, serving as a spatial reference for navigating between applications. (b) Snap: A flat NDH fixes the interaction window, enabling the DH to perform direct and precise interactions on the palm. (c) Over-Snap: A widely open NDH transitions the system to coarse-grained application management. Users can adjust window orientation via NDH rotation and relocate the window using DH pinch gestures. Both physical hands are remapped to the virtual window to mimic direct manipulation while allowing more comfortable postures.

Virtual Reality (VR) offers the potential for productivity work by providing expansive workspaces anywhere, yet supporting bare-hand interaction with multiple 2D windows remains challenging. We present HandPad, an asymmetric bimanual interaction framework that combines differentiated hand roles, self-haptic interaction, and hand remapping for multi-window interaction. The non-dominant hand sets a spatial frame, selects the interaction context, and provides a self-haptic surface, while the dominant hand performs on-hand touch

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and complementary mid-air manipulation relative to that frame. HandPad organizes interaction into three modes supporting cross-window navigation, within-window content manipulation, and application-window management. It also remaps the physical hands to virtual windows, allowing users to interact from more flexible physical postures regardless of window location. An exploratory study suggests that HandPad provides coherent interaction workflows, while revealing design tradeoffs between on-hand and mid-air interaction, informing the design of future bimanual interfaces for multi-window VR workspaces.

CCS Concepts: • **Human-centered computing** → **Gestural input**; **Virtual reality**.

Additional Key Words and Phrases: Gestures, bimanual interaction, input remapping, virtual reality

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1 Introduction

Virtual Reality (VR) offers the potential for productivity work by providing portable, customizable environments that are unconstrained by physical displays and surrounding environments [17, 47]. As head-mounted displays increasingly support multiple 2D application windows [17, 44], users need to perform diverse operations including manipulating virtual contents, transferring contents across windows, and arranging application layouts [6, 32, 53, 57]. Supporting these heterogeneous tasks using only bare-hand input remains challenging. While controllers and external devices such as mice or tablets provide familiar interaction techniques [15, 19], they introduce additional hardware and reduce the portability of mobile VR systems. Recent advances in optical hand tracking have therefore renewed interest in bare-hand interaction for productivity-oriented VR, as demonstrated by commercial headsets such as the Meta Quest 3 and Apple Vision Pro. However, designing a coherent interaction workflow that accommodates different levels of interaction and a wide range of tasks remains an open challenge.

Asymmetric bimanual interaction provides a promising design principle for addressing this challenge. In contrast to many current commercial VR systems, where both hands perform interchangeable functions, asymmetric interaction assigns complementary yet distinct roles to the dominant and non-dominant hands: the non-dominant hand (NDH) establishes the spatial frame and interaction context, while the dominant hand (DH) performs detailed manipulation relative to that reference [18, 24]. This division of labor has long been shown to support coordinated and efficient interaction in desktop and pen-based systems [5, 23, 26, 30, 54]. Modern VR headsets are well suited to this interaction model because they continuously track both hands and their spatial relationship, allowing the two hands to assume complementary rather than identical roles. We therefore investigate how asymmetric hand roles can organize interaction across multiple virtual windows using only bare-hand input.

To truly leverage the benefits of bimanual asymmetry [18], we propose HandPad, an asymmetric bimanual interaction framework for multi-window interaction in VR (Figure 1). HandPad assigns the NDH three complementary responsibilities: (1) establishing a **spatial interaction frame** (i.e., interaction window) and providing a reference to anchor the DH's motions; (2) switching between various **interaction contexts** through hand poses for different DH inputs; (3) providing a **self-haptic surface** to stabilize the reference frame for precise DH touch interactions. Within this context and reference frame provided by NDH, the DH performs on-hand touch and complementary mid-air manipulation. To accommodate the diverse requirements of multi-window interaction, HandPad organizes interaction into three modes: Semi-Snap, Snap, and Over-Snap. These modes

are designed for cross-window navigation, within-window content manipulation, and application-window management, respectively. In addition, HandPad spatially remaps the physical hands to virtual windows so interaction is not constrained by the windows' displayed locations [13, 40, 57], allowing users to interact using more comfortable and flexible postures.

We conducted an exploratory study to examine participants' experiences with the integrated HandPad workflow. Specifically, we wanted to understand how users perceive and experience HandPad's asymmetric bimanual interaction during multi-window VR workflows. We are also interested in what interaction strengths, breakdowns, and tradeoffs emerge within the HandPad workflow. Our findings show that participants generally reported positive perceptions of ownership, agency, learnability, and tactile feedback. Their feedback also revealed important tradeoffs. On-hand interaction was perceived as more suitable for sustained and multi-touch manipulation, while mid-air input was preferred for brief interactions such as selection. Participants also found switching between interaction contexts using the non-dominant hand to be intuitive and seamless, although it may introduce additional interaction costs. We discuss these findings as design considerations for future hand-centered multi-window interfaces.

2 Related Work

2.1 Window Interactions in VR

The rapid development and growing affordability of VR technology have opened new opportunities for productivity work [14, 49]. Unlike traditional desktop setups, VR headsets are portable and body-centric, enabling users to organize and interact with multiple applications within immersive, private, and extensible virtual environments anytime and anywhere [44]. Commercial headsets from Meta and Apple, such as Meta Quest 3 and Apple Vision Pro, already support flexible and portable workspaces that extend beyond the constraints of physical displays, offering users large and unlimited virtual screens. Despite VR's spatial affordances, 2D interfaces remain popular because they provide familiar layouts, efficient information organization, and cognitively lightweight interaction for detail-oriented work across diverse domains, including information tasks and creative design [17, 29].

Controllers, gaze, and in-air gestures are the predominant modalities for interacting with virtual windows, each offering distinct trade-offs for productivity-oriented tasks [49]. Specifically, controllers provide explicit buttons and accurate pose tracking [39], gaze enables rapid target acquisition [27, 43], and bare-hand mid-air input reduces reliance on external hardware [41, 51]. Researchers have also integrated conventional input devices, including mice and keyboards [19, 34] as well as styluses and tablets [6, 15, 48], into VR to leverage users' familiarity and established capabilities. However, these devices, originally designed for static physical displays, may not adapt well to dynamic virtual displays [49] and are not always practical for mobile or on-the-go use [44]. Meanwhile, multi-window workflows involve a diverse set of operations, including discrete selection, continuous content manipulation, cross-window transfer, and application-window management. This diversity makes it difficult for a single interaction technique to effectively support all tasks. Consequently, enabling versatile multi-window interaction in large virtual workspaces without relying on dedicated hardware remains a key challenge for mobile VR.

2.2 Hand-Centered Interaction

Hand-centered interaction offers a compelling alternative for mobile VR by eliminating the need for external devices while providing a more holistic experience that combines proprioception with natural manual interactions [3, 45].

2.2.1 Asymmetric Bimanual Interaction. When interacting with an object, users typically coordinate both hands in distinct roles, where the non-dominant hand holds the object for the dominant hand to interact with it. Guiard's Kinematic Chain model [18] explains that the non-dominant hand establishes the spatial frame and interaction context, while the dominant hand executes fine-grained actions relative to that established reference. This coordination leverages the benefits from human proprioception, as users can easily perceive the relative position of their hands [25, 26]. Such asymmetric bimanual interactions are essential for enhancing task performance of productivity tasks, particularly in terms of precision and speed [5, 24, 30].

However, standard touch interfaces often struggle to support asymmetric bimanual interaction because they lack interaction context, such as the ability to distinguish between the dominant and non-dominant hands, leading to fundamental role ambiguities [54]. Therefore, prior work has relied on external cameras to track both hands. For example, Imaginary Interfaces [20] allows users to define an interaction space by forming an L-shape with the NDH and performing drawing gestures with the DH within the defined frame. PalmRC [11] similarly leverages the user's palm as an interactive surface, enabling menu navigation through bare-hand input while maintaining visual focus on the display. Although modern VR headsets, such as the Meta Quest 3 and Apple Vision Pro, also support simultaneous tracking of both hands, current interaction techniques typically assign them interchangeable functions, overlooking the dexterity and specialized manual roles of the two hands.

2.2.2 On-Hand Interfaces. Another factor that limits the usability of bare-hand interaction for productivity work in VR is insufficient stability and precision, as well as the difficulty of controlling movements without physical support. These limitations can result in uncertain touch registration and reduced efficiency [2, 10, 38]. Prior work has addressed these challenges by introducing haptic feedback through daily physical objects [10, 56], digital tablets [6, 48], or customized wearable haptic devices [13, 31]. A lighter-weight alternative is self-haptics, which leverages the user's own body as a tactile medium [12, 35, 36, 45]. Hands are inherently available, low-cost, and seamlessly integrated with the human sensorimotor system through proprioception. Prior research has also demonstrated the versatility of the hand as an input surface. For example, the palm has been explored as a touchscreen-like surface [22, 58], a touchpad-like surface [52], and a menu selection tool [7, 11, 37, 46, 58], enabling users to focus on virtual content while benefiting from self-haptic feedback on the skin. In addition, touch interaction on the hand surface can leverage its physical constraints, mitigating accuracy issues caused by depth perception challenges in the 3D space. More recently, Handows [53] demonstrated that the non-dominant palm can serve as a miniature interface for managing multiple virtual windows through smartphone-inspired gestures.

Despite these advances, reliable touch detection on the hand without external sensing hardware remains challenging. Moreover, leveraging hand-based interfaces for productivity tasks that involve large displays, multiple windows, and prolonged engagement introduces additional constraints. While Handows [53] demonstrates palm-based interaction for managing multiple virtual windows, it focuses on window-level operations rather than interaction with content within and across windows. In addition, the small on-hand interaction space, potential instability of the hand, and ergonomic concerns during extended use further challenge the design of hand-centered interfaces that support integrated multi-window workflows.

2.3 Input Remapping

The "gorilla-arm effect" is a well-known limitation of bare-hand interaction in VR [4, 21]. Although direct hand manipulation is intuitive and widely adopted, it often requires users to extend their

arms to reach virtual windows positioned high or at a distance, which is common in productivity-oriented VR environments. Input remapping offers a solution by decoupling the physical motion of the user's hands from the virtual interaction space [28, 55]. This approach enables users to perform gestures at a lower, more relaxed position while their inputs are spatially mapped to eye-level or distant virtual windows. Such techniques leverage proprioception to preserve a sense of direct manipulation, even when physical and virtual hand positions are not spatially aligned. For example, ARPads [8], Virtual Pads [1] and Redirected Pinch [57] introduce virtual interaction planes near the user's body, allowing hand movements on the plane to indirectly control a cursor on a 2D window. Ownershift similarly allows users to gradually lower their hands from elevated window positions to achieve more ergonomic interaction [13].

3 HandPad Design

HandPad is a bare-hand interface that leverages the user's NDH to establish a mobile spatial frame and interaction context. Defined relative to that established reference, the DH performs touch interactions on the NDH with self-haptic feedback or mid-air gestures to modulate the frame. Through asymmetric bimanual coordination, the NDH further employs distinct gestures to enable seamless mode switching. Overall, HandPad supports a suite of interaction techniques that operate across three levels of components: the interaction window, which defines the spatial frame; the virtual content within the window; and the application window.

3.1 Design Objectives

While bare-hand mid-air interaction is widely adopted in commercial VR headsets due to its intuitiveness and availability, interacting with hand interfaces can introduce several challenges. We derived the following design objectives from prior work on asymmetric bimanual interaction, hand-centered input, self-haptics, input remapping, and multi-window interaction in VR, together with the interaction requirements of productivity-oriented workflows involving content manipulation and window management.

Differentiated Hand Roles. Previous work has shown the benefit of bimanual asymmetry [5, 18, 24, 30] and haptic feedback [2, 12, 50] for enhancing the interaction efficiency in VR. We assigned different roles to the two hands. The NDH serves as (1) a spatial frame to establish the interaction window and provide a reference to anchor the DH's motions; (2) a tangible surface stabilizing the frame to support precise DH touch interactions on the NDH with self-haptics; (3) a kinesthetic mode-switching modality using gesture to phrase interactions and create context-specific meanings for different inputs such as manipulating contents or windows.

Posture Flexibility. Directly interacting with virtual content using hands in VR may cause fatigue and discomfort for users due to the gorilla-arm effect [21]. Especially for prolonged productivity tasks, users need to perform interactions with a comfortable posture. Inspired by previous work that created a virtual touchpad and allowed users to interact with it at a lower and more comfortable location [1, 8], we employ the hand as a touchpad. Users can perform gestures at a lower and more comfortable posture. Meanwhile, both hands and their interactions will be remapped to the corresponding virtual window, creating the sensation that the user is directly touching the content.

Hand-Referenced Input. Another limitation of hand-based interfaces in VR is users' limited ability to maintain stable hand positions in mid-air, particularly during 2D window interaction. Even minor tremors can degrade performance in simple tasks such as selection. To address this issue, we introduce a snapping mechanism that anchors the interaction window to the application window, thereby stabilizing the spatial reference frame. This allows the dominant hand (DH) to perform

touch interactions or mid-air gestures relative to a fixed frame, reducing the impact of unintended hand jitter.

Compatibility with Direct Mid-Air Input. Designed gestures should not conflict with users' natural resting postures or commonly adopted mid-air gestures, such as pinching and grabbing in current commercial VR headsets. We therefore assign uncommon yet easy-to-perform hand postures to the NDH for specific functions, such as mode switching, while preserving familiar DH gestures, including pointing, pinching, and multi-touch input.

Coverage Across Interaction Levels. Beyond basic interactions such as clicking, dragging, and scrolling, hand-based interfaces must support the complex and compound manipulations inherent to productivity work. Productivity tasks frequently require coordinated multi-touch operations, including zooming and rotating content, transferring information across multiple windows, and managing application windows through moving, resizing, and rotating.

Adaptability to Window Position and Scale. Although the physical hand provides only a limited interaction surface, it must flexibly accommodate application windows of varying sizes. As proposed, the hand serves as a spatial reference frame that instantiates an interaction window on top of application windows. This interaction window should be resizable and seamlessly movable across multiple applications through coordinated hand gestures, enabling fluid transitions between contexts. Furthermore, when users need to interact with content beyond the bounds of the interaction window, the hand can function as a stable anchor that supports an effectively unbounded virtual interaction space around it. This allows users to extend their input beyond the physical hand surface and perform mid-air interactions, such as clicking, while preserving spatial continuity and allowing interaction with flexible postures.

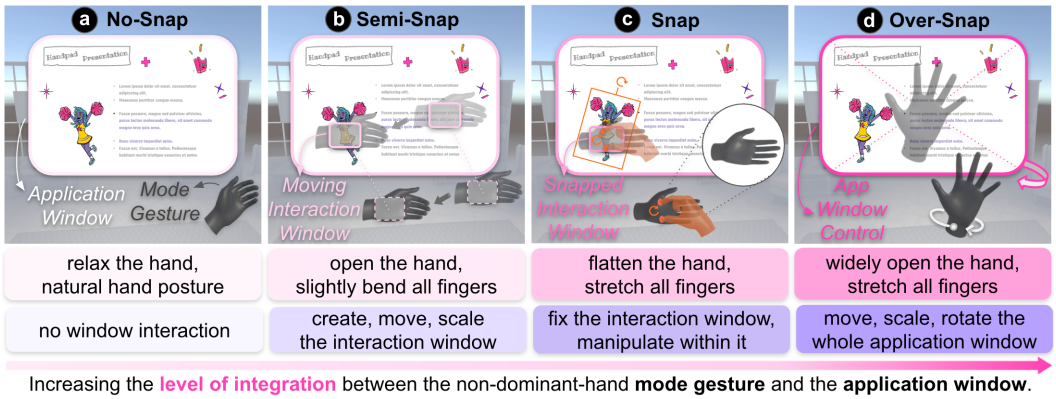


Fig. 2. Non-dominant hand gestures set different interaction modes. (a) No-Snap: Natural hands, no window interaction. (b) Semi-Snap: Bend the NDH to create, move, and scale the interaction window. (c) Snap: Fix the interaction window and manipulate 2D content via DH gestures on the flat NDH. (d) Over-Snap: Use a widely open NDH to move, scale, and rotate the entire application window. Dark gray hands indicate the physical NDH; orange hands indicate the physical DH; translucent hands indicate the remapped hands on the window.

Table 1. Joint configurations across different NDH snap modes.

	No-Snap	Semi-Snap	Snap	Over-Snap
 **Joint flexion** angle	200–250°	80–150°	40–60°	< 40°
Inter-finger angle	30–40°	30–40°	30–40°	> 60°

3.2 Interaction Modes and Gesture Recognition

HandPad is designed to address the interaction objectives outlined above. It employs the user's NDH to establish a mobile spatial frame and interaction context, enabling fluid mode switching while allowing the DH to perform fine-grained manipulations relative to this reference frame.

3.2.1 Non-Dominant Hand Mode Gestures. HandPad enables the NDH to interact with the application window through four progressively window-integrated modes: (1) **No-Snap**, (2) **Semi-Snap**, (3) **Snap**, and (4) **Over-Snap** as shown in Figure 2. Users transition between modes by performing distinct NDH gestures. Specifically, changes in finger flexion angles and inter-finger spreading angles are used to differentiate modes, and increasing finger stretchiness corresponds to a higher level of integration with the application window.

In No-Snap (Figure 2a), the NDH remains relaxed with minimal finger extension, resulting in no window interaction. In Semi-Snap (Figure 2b), users slightly bend the NDH to create, move, and scale the interaction window, thereby establishing the interaction frame. When users stretch their fingers to form a flat surface, the system enters Snap mode (Figure 2c), which fixes the interaction window and enables manipulation of 2D content via DH gestures on the flat NDH. Finally, in Over-Snap (Figure 2d), users fully spread their fingers, triggering the highest level of window-integration and allowing manipulation of the entire application window, including moving, scaling, and rotating.

To ensure that NDH gestures do not conflict with users' natural resting postures or commonly adopted mid-air gestures, such as pinching and grabbing in commercial VR headsets, we conducted an informal study with four participants. Participants were asked to rest their hands naturally and perform common mid-air gestures while we recorded joint flexion and inter-finger angles. We observed that, during these activities, the total joint flexion angle exceeded 250°, while the total inter-finger angle remained below 30°.

To minimize interference with everyday gestures, we designed HandPad NDH gestures to occupy a distinct range of hand configurations, with total joint flexion angles (JFA) between 40° and 250° and total inter-finger angles (IFA) between 30° and 60°. As summarized in Table 1, No-Snap occurs when JFA is within 150°–250° and IFA is within 30°–40°. Semi-Snap is triggered when JFA decreases to 80°–150°. Snap occurs when JFA further decreases to 40°–60°. Finally, Over-Snap is activated when the hand becomes widely opened, characterized by JFA < 40° and IFA > 60°. These thresholds were determined from the informal study described above and were subsequently used consistently across participants in the exploratory evaluation.

3.2.2 Dominant Hand Manipulation Gestures. HandPad leverages the DH to perform touch gestures on the NDH palm, supporting both single-touch and multi-touch interactions, as well as mid-air gestures anchored to the NDH's spatial position for diverse window manipulations (Figure 3). In the following, we describe the touch detection method used to identify DH–NDH contact and the gesture recognition mechanism implemented through a state-machine-based approach.

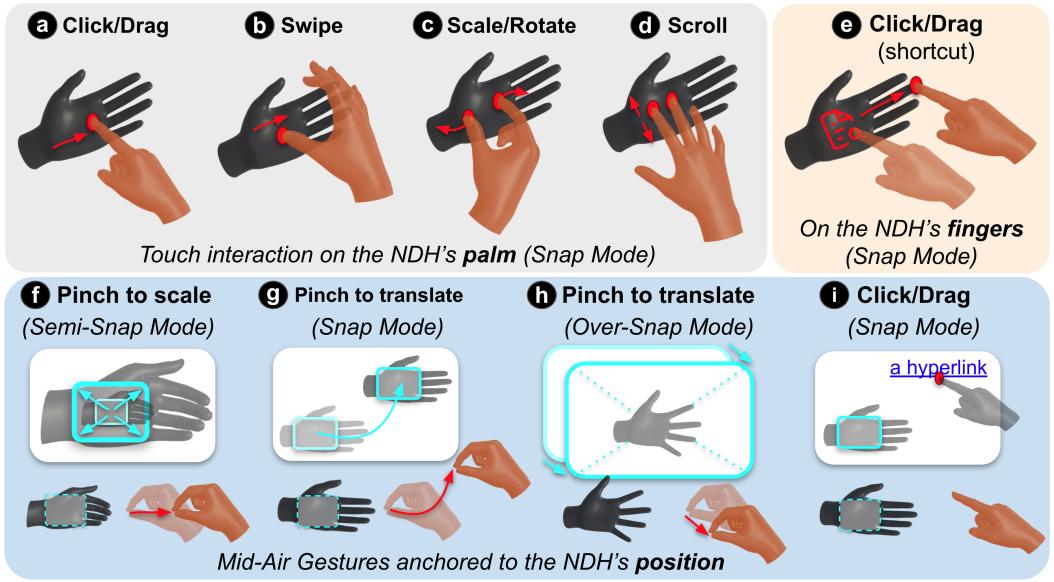


Fig. 3. Dominant-hand manipulation gestures. Snap mode enables palm-based touch interactions for (a) click/drag, (b) swipe navigation, (c) scale/rotate, and (d) scroll; finger touch supports (e) shortcut operations. NDH-anchored mid-air gestures include (f) pinch to scale (Semi-Snap), (g) pinch to translate the interaction window (Snap), (h) pinch to translate the application window (Over-Snap), and (i) click/drag when content lies beyond the hand surface.

Touch Detection. We developed a real-time vision-based touch detection model to identify both single-point and multi-point contacts between the DH and the NDH using the built-in hand tracking of the Quest headset. To reduce tracking jitter, we applied a 1€ filter [9] to hand and finger motion, using parameters $f_{c\,\min} = 0.9$ and $\beta = 90$. We represented both hands as geometric proxies constructed from the tracked hand representation in Unity. The touch classifier used the fingertip-to-surface distance and the potential contact location on the surface as input features. Specifically, the fingertip-to-surface distance is computed from the DH fingertip to the NDH proxy along the local surface normal. A ray cast from the DH fingertip to the proxy determines the potential contact location. The resulting intersection point is then transformed into the local coordinate system of the NDH proxy and normalized with respect to the proxy bounds.

We collected 5,000 touch samples by tapping different regions of the NDH with the DH index fingertip and 5,000 no-touch samples by hovering above the same regions. Each sample consisted of the fingertip-to-surface distance d and the normalized (x, y) coordinates of the ray-cast intersection. The dataset was split evenly between independent training and evaluation sets. We trained a random forest classifier with $N = 100$ trees and a maximum depth of 25. The resulting model achieved an accuracy of 89.3%, with a false-negative rate of 3.2% and a false-positive rate of 7.5%. We then selected an operating confidence threshold that prioritized avoiding missed contact, reducing the false-negative rate to 1.3% while increasing the false-positive rate to 12.2%. This tradeoff was intentional because missed touches were more disruptive to interaction continuity than early near-surface detections. However, the increased false-positive rate may reduce touch accuracy during prolonged use, particularly when users hover near the NDH surface without intending

to touch it. Accordingly, participants were instructed to avoid intentionally hovering their DH fingertip close to the NDH surface unless they intended to initiate touch.

Manipulation Gesture Recognition. Once contact between a DH manipulation fingertip and the NDH palm surface is detected, the touch is classified into three states: touched, dwelling, and moving, as shown in Figure 8, following prior touchpad-based approaches [33]. Specifically, a touch enters the dwelling state if the contact point remains within the same location for longer than 450 ms. A touch or dwelling enters the moving state if the displacement of the contact point exceeds 10 mm. We combine these touch states with the identity of the contacting fingers to distinguish gestures such as clicking, dragging, swiping, scrolling, scaling, and rotation. Detailed state representations and gesture sequences are provided in Appendix A. Mid-air click and drag gestures are detected through fingertip collision with the virtual window, while pinch gestures are recognized from the distance between the thumb and index fingertips.

3.3 Multi-Level Window Interaction Techniques

By combining NDH-enabled interaction modes (Figure 2) with DH manipulation gestures (Figure 3), we establish a spatial reference frame using an interaction window created by the NDH, allowing both hands to perform manipulations relative to this frame. Based on this design, we develop interaction techniques that operate at three levels: (1) across-window interaction in Semi-Snap, (2) within-window content interaction in Snap, and (3) application-window management in Over-Snap, reflecting an increasing level of integration between the NDH and the application window. We also support complementary mid-air gestures across these interaction contexts.

3.3.1 Interaction Window. The interaction window defines the active interactive area on the application window. It is a floating window created and manipulated under the **Semi-Snap** mode. Users create the interaction window by gazing at or pointing the NDH toward the desired location while performing the Semi-Snap gesture (Figure 2b). To accommodate large application windows and support seamless transitions, the interaction window can be relocated and resized dynamically.

Users can relocate the interaction window by gazing at or pointing to another location while remaining in Semi-Snap mode. If **Snap** mode has already been triggered, users can also reposition the interaction window by performing a pinch gesture with the DH and dragging it around the NDH, as shown in Figure 3g.

To resize the interaction window, users can perform a pinch gesture with the DH (Figure 3f). Moving the pinched fingers toward the NDH reduces the size of the interaction window, while moving them away enlarges it. The size of the interaction window influences cursor behavior on the application window because interactions on the NDH are mapped one-to-one to the interaction window. A larger interaction window produces a higher control-to-display (C/D) gain, enabling faster navigation and large-scale manipulations such as scrolling quickly or interacting with large images. Conversely, a smaller interaction window supports finer control for tasks such as signing or annotation.

3.3.2 Across-window Interaction. During **Semi-Snap** (Figure 2b), the interaction window can be translated, which is particularly useful for large-window and across-window interactions, where long-distance manipulations on the NDH palm would otherwise require repeated small drags. For example, to move an image from the bottom of an application window to the top, or even across different application windows (Figure 4), the user can long-press on the NDH with the DH to select the image. The selected image is then attached to the interaction window and moves with it as the NDH relocates the window to the desired location. The user completes the operation by lifting the DH fingertip from the NDH and dropping the image at the target location.

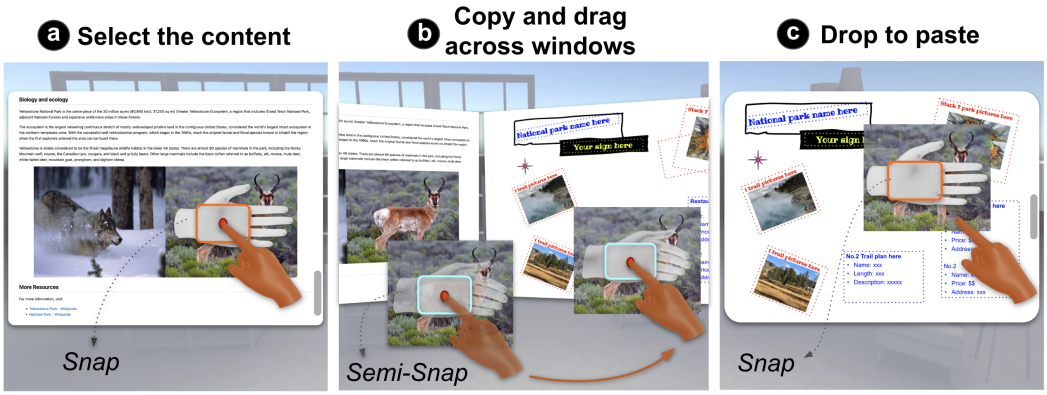


Fig. 4. Across-window drag-and-drop interaction. The user first selects content on the source window through touch interaction on the NDH. While maintaining the selection, the interaction window and selected content move with the NDH toward the target application window. Releasing the DH touch drops the content at the target location. White hands indicate the virtual NDH and orange hands indicate the virtual DH.

3.3.3 Within-Window Content Interaction. When users interact with 2D content on an application window (e.g., text, images, or buttons), the interaction window can be fixed over the content using the **Snap** gesture, as shown in Figure 2c. Once the interaction window is stabilized, users can relax the NDH into a more comfortable posture while performing fine-grained interactions on it with the DH. Based on the interaction contexts established by the NDH during **Snap** mode, three types of interactions are enabled: (1) touch interactions on the NDH palm, (2) touch interactions on the NDH fingers, and (3) mid-air gestures anchored to the NDH position.

Touch interactions on the NDH palm. While controllers and mid-air gestures support only a limited set of interactions due to the lack of physical surface support and the under-utilization of finger dexterity, HandPad leverages the tangible flat surface of the palm while allowing both hands to move freely. Because the palm area is comparable in size to a touchpad, HandPad supports classic single-touch and multi-touch interactions similar to those on a touchpad.

For example, users can repeatedly click and drag on the NDH palm with any DH finger to control a cursor or perform stroke-based input (Figure 3a). To navigate across a webpage, users can swipe the DH thumb on the NDH palm to move up, down, left, or right (Figure 3b). For precise manipulation of 2D content, such as scaling or rotating text or images, users can use the DH thumb and index finger to perform fine multi-touch adjustments on the NDH palm (Figure 3c). To scroll through a webpage, users can perform brief two-finger swipes with the DH index and middle fingers on the NDH palm (Figure 3d). These touch interactions operate under relative control and incorporate inertial effects based on the speed of finger movement. For instance, faster swipes on the NDH palm produce longer scrolling distances, which is particularly beneficial when navigating large windows.

Touch interactions on the NDH fingers. Beyond the NDH palm surface, fingers can also be leveraged for efficient touch interactions, such as shortcuts [7], since users can easily locate each finger through proprioception. As an example scenario shown in Figure 3e and Figure 5, HandPad enables a temporary clipboard that allows users to quickly store multiple 2D contents (e.g., text or images) or files across different NDH fingers. Users first select the desired content on the NDH palm and then move it to a specific fingertip using a DH manipulation finger, effectively storing the item on

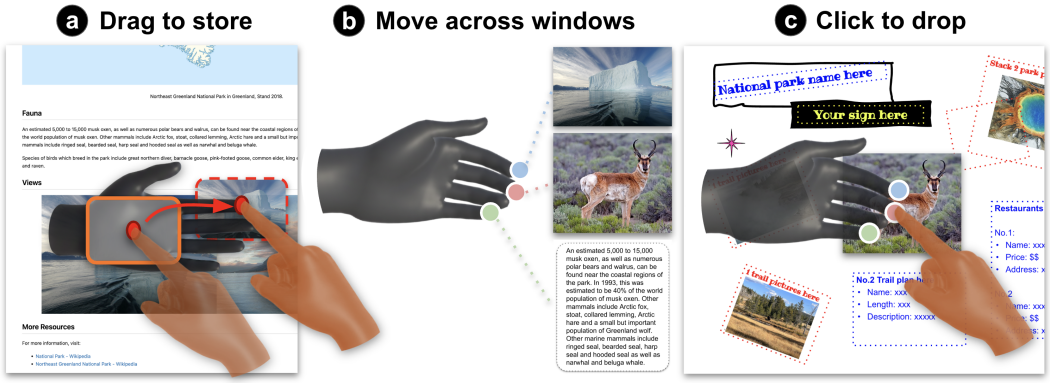


Fig. 5. Example finger-based shortcut interaction for a temporary clipboard. Selected content can be dragged from the NDH palm to an NDH fingertip for temporary storage. The stored content moves with the NDH across application windows and can later be retrieved by touching the corresponding fingertip.

that finger. The stored contents then move together with the NDH, allowing users to transport them across multiple application windows. To retrieve an item, users simply click on the corresponding NDH fingertip and drag it back to the NDH palm, placing the stored content at the desired target location.

Mid-air gestures anchored to the NDH position. While the NDH palm supports diverse and fine-grained touch interactions similar to a touchpad, application windows in VR can be extremely large compared to the small surface area of the hand. In scenarios where the interaction window is fixed by the NDH, users may need to perform simple, quick, and repetitive actions outside this window, such as clicking on items beyond the interaction area. Repeatedly switching NDH modes or relocating the interaction window for these minor interactions can become tedious. In contrast, mid-air gestures are well-suited for such lightweight interactions without physical surface or haptic feedback. HandPad therefore allows the NDH to serve as an anchor while enabling an unlimited virtual interaction space around the interaction window. This allows users to directly point and interact with the application window in mid-air using the DH, such as selecting hyperlinks (Figure 3i).

3.3.4 Application-Window Management. Efficient manipulation of entire application windows, such as relocating, minimizing or opening, rotating, and resizing, is essential for managing multiple windows in VR. HandPad enables these operations through the **Over-Snap** gesture performed by the NDH (Figure 2d), which snaps the entire application window to the NDH for direct manipulation. To relocate an application window, users can point at the target application window and perform the Over-Snap gesture using NDH. The application window then attaches to the NDH, allowing users to move it to a new location before releasing the gesture to place it. However, given the large 3D virtual space, moving windows solely with the NDH may be inefficient when the target location lies beyond arm's reach. To address this limitation, HandPad allows the DH to perform relative control using pinch gestures (Figure 3h). By pinching relative to the NDH, users can translate the application window along the x, y, and z axes. Window orientation can be adjusted by rotating the NDH through wrist twisting, which allows the window to face different directions. Users can also resize the application window by performing a scale gesture with the DH thumb and index finger on the NDH during Over-Snap mode.

3.4 Hand Remapping

When users initiate HandPad interaction, both physical hands are translationally and rotationally remapped to the application window to enable direct-like manipulation. Since the physical NDH serves as the spatial reference frame represented by the interaction window, we retarget the physical NDH to the location of the interaction window and render a corresponding virtual NDH on the application window. The physical DH is then retargeted according to the relative position and orientation between the physical and virtual NDH (Figure 2b–d). After hand remapping, users can relax their physical hands while the virtual hands remain aligned with the application window.

4 HandPad Exploratory Study

Prior research has highlighted the advantages of asymmetric bimanual interaction [25, 26] and hand-based interfaces that provide self-haptic feedback and tangible manipulation [12, 45]. Building on these insights, we conducted an exploratory study to examine how participants experienced HandPad as an integrated interaction workflow in representative multi-window scenarios.

4.1 Apparatus

Our study involved web applications displayed across multiple windows. The applications were developed and hosted on a Python Flask server. Application windows were rendered in VR using a Unity web browser plugin by Vuplex. HandPad interactions performed on windows in VR were transmitted to the web applications through a custom event sender–listener system, enabling real-time interaction. Hand tracking, touch interaction, and mid-air gesture recognition were implemented using the Meta XR Interaction SDK in Unity. The study application ran on a laptop equipped with an Intel Core i7 processor and an NVIDIA RTX 3080 GPU.

4.2 Participants

Nine participants (6 male and 3 female), aged 19–29 years, all right-handed, were recruited from the university. Their experience with VR varied. Two participants reported frequent use (daily or weekly), while the others used VR occasionally for gaming, product exploration, or participation in research studies. Each study session lasted approximately 60 minutes, and participants received $20 USD in compensation. The study followed the ethics review requirements of the authors' research institution and was approved by the institutional review board.

4.3 Procedure

Participants were seated and wore a Meta Quest Pro VR headset throughout the study. The study began with a practice session to familiarize participants with the HandPad system, including the four NDH snap modes and DH manipulation gestures for different types of window interactions: (1) interaction window manipulation, (2) interaction on a window, (3) window management, and (4) complementary mid-air gestures. Participants practiced these interactions by following the experimenter's instructions on a tutorial window until they felt comfortable using the system. The practice session lasted approximately 5–10 minutes. Afterward, participants used HandPad in two VR scenarios: (1) an open-book quiz and (2) a poster design task, each requiring interactions across three to four windows. During the tasks, the experimenter provided verbal guidance on the next steps and suggested interactions relevant to each scenario. After completing both scenarios, participants filled out a survey evaluating their experience with each HandPad interaction technique. Finally, they completed a post-study questionnaire and participated in a five-minute semi-structured interview to provide general feedback on using HandPad for productivity work in VR.

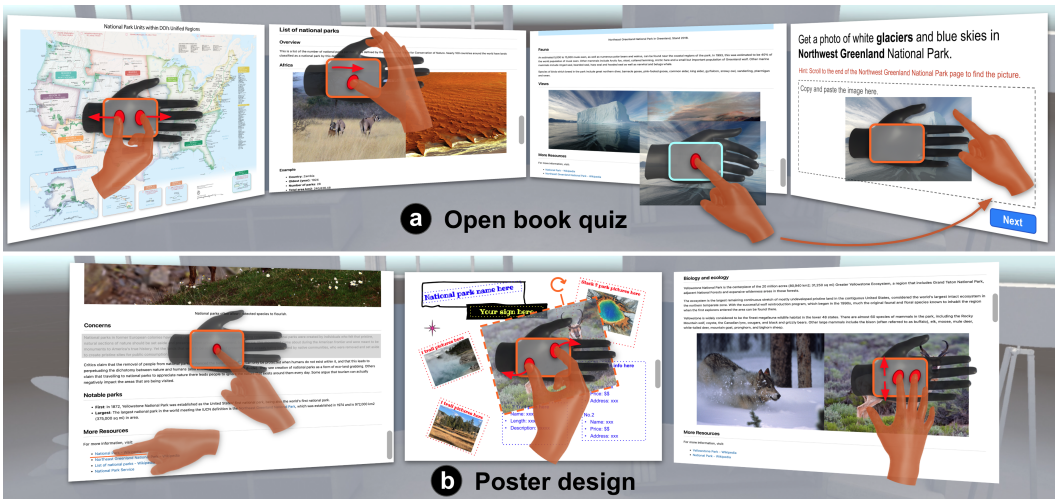


Fig. 6. Study scenarios used to evaluate HandPad in multi-window workflows. (a) Open-book quiz: participants searched for information across multiple application windows, navigated webpages and a map, and transferred text or images to answer questions. (b) Poster design: participants transferred content from source windows to a central canvas and performed fine-grained manipulations such as translation, scaling, rotation, drawing, and signing.

4.4 Tasks

Participants interacted with the HandPad system in two productivity-oriented tasks: (1) completing an open-book quiz and (2) designing an illustrated poster. We selected these two tasks to expose participants to complementary interaction demands representative of multi-window productivity workflows. The open-book quiz emphasized information search, navigation, and transferring content across multiple windows, while the poster design task emphasized sustained within-window manipulation, including dragging, scaling, rotating, and stroke-based input. Together, the tasks allowed participants to experience HandPad across cross-window interaction, within-window content manipulation, and application-window management.

Open Book Quiz. The quiz task involved four application windows, as shown in Figure 6a. One application window displayed the quiz questions, while the other three presented information sources from which participants could find the answers. These sources included (1) a map that participants needed to zoom and pan, and (2) two Wikipedia pages that participants navigated to locate relevant text and image information. After finding the required information, participants either clicked on the question page to enter an answer or copied content by dragging text or images from the information pages into the answer box. They then clicked a button to proceed to the next question. This quiz scenario was designed to allow participants to experience frequent information search and manipulation tasks using the HandPad interaction techniques.

Poster Design. The poster design task involved three application windows (Figure 6b). The center window served as the main canvas where participants created a poster, with visual guides such as text-box outlines, image placeholders, and a designated signature area. The left and right windows provided text and image materials for the design. Participants first searched for appropriate text and images, then dragged them onto the center design canvas. After placing the content, they performed fine-grained adjustments such as translation, rotation, and scaling to precisely position

the 2D elements within the required layout. Finally, participants signed their initials and drew additional shapes to personalize their poster. This design scenario was selected to allow participants to extensively explore fine-grained multi-touch interactions as well as precise cursor and stroke input using the HandPad techniques.

4.5 Data Analysis

We report descriptive statistics for the questionnaire data. To analyze participants' open-ended responses and post-study interviews, one researcher first reviewed the qualitative data and assigned descriptive codes to statements about participants' experiences with the interaction modes, on-hand touch, mid-air interaction, window manipulation, and hand remapping. The resulting codes and higher-level themes were then discussed and refined with a second researcher. Through this process, related codes were grouped into themes that captured recurring patterns across participants. These themes were used to organize and synthesize the qualitative findings.

4.6 Results

We first report participants' perceptions of ownership, agency, and tactile sensation [16], followed by perceived ease of use for the individual HandPad interaction techniques. We then report qualitative themes about HandPad interaction strengths, breakdowns, and tradeoffs that emerged from participants' open-ended responses and post-study interviews.

4.6.1 Ownership, Agency, and Tactile Sensation. HandPad remaps both hands to interact with large and distant windows while leveraging the NDH as a tangible surface to support comfortable interactions. However, this remapping may influence participants' perceptions of ownership and agency over the remapped hands. In addition, it is unclear whether touching the palm of the hand can produce a convincing tactile sensation when interacting with a virtual window. To examine these aspects, we selected relevant questions from the embodiment questionnaire [16], including Q1–2 for Ownership, Q6–9 for Agency, and Q10–13 for Tactile Sensation. Responses were collected using a 5-point Likert scale ranging from "strongly disagree (-2)" to "strongly agree (+2)".

Participants generally reported positive ratings for ownership (Median = 1, IQR = 1) and agency (Median = 2, IQR = 1) over their remapped virtual hands. Participants also rated positively the tactile sensation experienced when interacting with virtual windows through the NDH surface (Median = 2, IQR = 1).

Six participants emphasized the sense of control provided by the technique. For example, P7 noted, "*I felt more in control of what was happening with the HandPad.*" Similarly, P9 stated, "*HandPad gave me a sense of control over the window and the stuff I was selecting.*" In addition, all participants appreciated the tactile feedback provided by the hand surface when performing gestures that used the hand as a touch interface. For instance, P4 commented, "*It actually felt like I was touching something when I tapped on my palm, which made the interaction more natural.*"

4.6.2 Perceived Ease of Use. Participants generally found the HandPad interaction techniques to be learnable and practical, but their feedback on the ease of use of individual features varied (Figure 7). Responses were collected using a 5-point Likert scale from "very difficult (1)" to "very easy (5)".

Interaction Window. Participants' ratings were generally favorable for resizing and locating the active interaction area. In the **Semi-Snap** mode, participants found it easy to resize the interaction window (Median = 4, IQR = 1) and to locate the active interaction area using the interaction window (Median = 4, IQR = 2). Five participants appreciated the flexibility of adjusting the interaction window to accommodate large or distant windows. As P5 noted, "*The pad size can be adjusted and placed everywhere. Besides, the pad can be fixed or moved according to my demand.*"

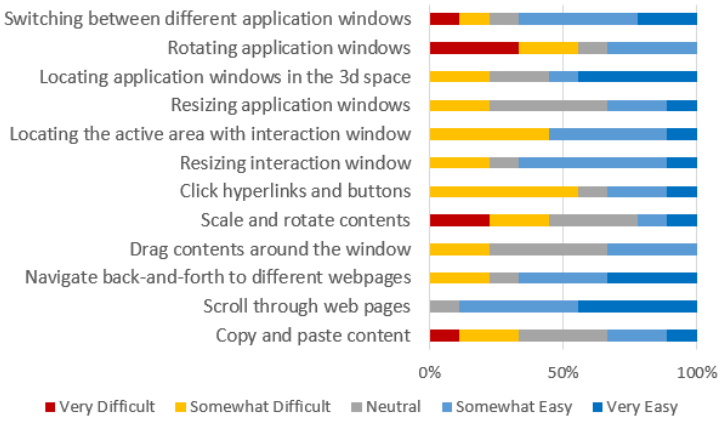


Fig. 7. Ease of Use for content, interaction window, and application window interactions.

Touch Interactions on a Window. In the **Snap** mode, participants found it easy to perform common interactions on a window, including single-touch interactions such as navigating back and forth between webpages (Median = 4, IQR = 2) and dragging content within a window (Median = 3, IQR = 1). Multi-touch interactions were also rated positively, including scrolling through webpages (Median = 4, IQR = 1) and performing multi-touch manipulations such as scaling and rotating 2D content (Median = 3, IQR = 1), which "*require some effort to achieve the correct position and orientation*" (P7). In the **Semi-Snap** mode, copying and pasting 2D content by dragging and dropping it across multiple windows was also rated as acceptable (Median = 3, IQR = 2). Surprisingly, five participants reported some difficulty when clicking hyperlinks and buttons (Median = 2, IQR = 2), especially when compared with mid-air selection. Two participants mentioned that using the Semi-Snap mode to locate the interaction area and then switching to Snap mode to perform the selection introduced "additional steps" (P3, P6). These responses suggest a tradeoff: on-hand interaction supported richer continuous manipulation, but the Semi-Snap-to-Snap sequence introduced overhead for brief selections.

Window Management. Regarding manipulation and management of application windows in the **Over-Snap** mode, locating application windows in 3D space received a median easiness rating of 4 (IQR = 2), resizing application windows received a median of 3 (IQR = 1), and switching between different application windows received a median of 4 (IQR = 1). However, participants encountered some difficulties when rotating application windows (Median = 2, IQR = 3). As P1 noted, this may be attributed to the high dexterity of hand motion, which introduces "*too much degree of freedom for [window] rotation*" and makes it harder to control precisely.

4.6.3 Interaction Strengths and Breakdowns.

Bimanual Interactions. Seven participants appreciated that using the NDH to establish a spatial frame and enable different interaction contexts through Snap mode gestures was intuitive, distinguishable, and easy to learn. They noted that the mode gestures were easy to perform and distinguish from one another. Participants also appreciated the Semi-Snap mode for positioning the interaction window and the Snap mode for stabilizing the interaction. As P1 noted, "*It helped me to switch between the active and focused interaction area on the large window.*" Participants further valued the seamless transition between interaction modes. For example, P7 mentioned, "*I like that*

the gesture (Semi-Snap) is a step before the Snap because it is very easy to go from Semi-Snap to Snap and come back." Four participants also enjoyed being able to touch the NDH with the DH "*just like a touchpad*" (P4), which provided a tangible reference during interaction. Regarding window management, six participants appreciated the application window resizing and repositioning features provided by the Over-Snap mode. For instance, P9 noted that "*it allowed me to reposition the window closer to my eyes.*" However, dragging application windows across long distances by only moving the NDH could be somewhat challenging and "*needed a lot of attempts*" (P7). In these cases, participants also used the complementary DH mid-air gestures anchored to the NDH to manipulate the window over a larger range.

Hand Remapping. Eight participants appreciated that HandPad allowed them to rest or lower their hands during interaction while interacting with the virtual window. For example, P5 reported, "*With the Snap mode, I can rest my arm, so it's more comfortable for me to operate.*" However, participants may require practice or reminders to fully leverage this posture flexibility. Two participants mentioned that they would forget to relax their hands when they became highly engaged in the task, only noticing the strain afterward. For instance, P4 mentioned, "*It mitigates physical exertion if I remember that I can freely use my hand, but I sometimes forgot that.*" On the other hand, if participants lowered their hands too much, they could fall outside the camera's tracking range, which might interrupt the interaction flow. This sometimes resulted in repeated attempts. As P2 explained, "*Sometimes [tracking] is not accurate, so I need to do the same thing multiple times.*" Thus, hand remapping expanded the range of possible physical postures, but its perceived value depended on participants remembering to reposition their hands and remaining within the headset's tracking volume.

Participants' Reflections on Prior Controller and Mid-Air Interaction. During the post-study interviews, all participants were invited to reflect on HandPad relative to their prior experiences with controller-based and mid-air interaction. Three participants noted that the hand remapping made the interaction feel more natural and intuitive than controller-based input. For example, P2 stated, "*I like how it imitates my hand movement, and it's great to do activities on a big screen without controllers. It somehow feels like I am using a touch screen with feedback…*" Participants (4/9) who were familiar with interacting with 2D windows using mid-air gestures commented positively on the multi-touch and cross-window interactions supported by HandPad. They also noted that these interactions felt more fluid and easier to control than traditional mid-air gestures, which often lack a stable physical reference.

5 Discussion and Design Implications

Our exploratory study examined how participants experienced HandPad as an integrated interaction framework for multi-window workflows in VR. Participants generally reported positive perceptions of ownership, agency, and tactile correspondence, while also identifying tradeoffs associated with mode transitions, window manipulation, and hand remapping. We discuss these findings below to derive design considerations for future hand-centered interfaces that combine asymmetric hand roles, hand-centered interaction, and complementary input modalities.

5.1 Contextual Hand Roles for Multi-Level Interaction

HandPad assigns the non-dominant hand to establish both the spatial reference frame and the interaction context, while the dominant hand performs manipulation relative to that reference. This design is consistent with prior work on asymmetric bimanual interaction [18, 24]. Participants generally appreciated the clear separation between the Semi-Snap, Snap, and Over-Snap modes and reported that the gestures were easy to distinguish after training. The layered interaction structure

allowed users to progressively transition from across-window navigation to content manipulation and to application-window management without changing interaction paradigms.

HandPad extends prior palm-based window-management approaches such as Handows [53], which use the non-dominant palm to control application windows, by using the NDH as a persistent contextual reference across window management, within-window content manipulation, and cross-window interaction.

Meanwhile, participants noted that entering Snap mode before performing simple actions such as clicking introduced additional interaction steps. This observation suggests that contextual hand roles are particularly valuable when users remain within an interaction context long enough to offset the cost of entering it. Future hand-centered interfaces may therefore benefit from assigning the non-dominant hand persistent contextual responsibilities while allowing lightweight or opportunistic interactions to remain directly accessible without requiring a complete mode transition.

5.2 On-Hand Touch Interaction

Participants consistently reported that touching the non-dominant hand surface felt similar to using a touchpad and appreciated the presence of physical contact during clicking, dragging, scrolling, and multi-touch manipulation. The non-dominant hand simultaneously serves as a tangible interaction surface and a spatial reference frame, allowing the dominant hand to perform touch interactions relative to a stable hand-centered coordinate system. This observation aligns with prior work showing that physical constraints can improve interaction stability by reducing unintended movement [2]. Our findings also extend prior on-hand and self-haptic interfaces [12, 45, 52] by showing how the hand surface can serve not only as a localized input region, but as part of a broader multi-window interaction workflow involving versatile content manipulation.

Interestingly, participants particularly valued on-hand interaction for sustained manipulation tasks, including scrolling, dragging, and multi-touch operations. These findings suggest that hand-centered touch interaction is especially appropriate for interactions that benefit from continuous physical contact and stable relative control.

5.3 Complementary Roles of On-Hand and Mid-Air Input

Although participants generally appreciated on-hand interaction, they also preferred direct mid-air input for some lightweight operations. Two participants reported that switching from Semi-Snap to Snap before clicking introduced unnecessary overhead for brief interactions, while mid-air pointing allowed them to complete these actions more directly. This tradeoff is consistent with prior work showing the accessibility of mid-air hand input for direct spatial interaction [41, 51], while surface-supported interaction can provide a stable physical reference for more sustained or precise manipulation [2, 10]. Our findings suggest that these interaction styles can support different phases of the same multi-window workflow.

Rather than viewing on-hand interaction and mid-air interaction as competing techniques, our findings suggest that they serve complementary roles within the same workflow. On-hand interaction appears particularly suitable for sustained manipulation, multi-touch input, and interactions that benefit from passive haptic feedback and relative control. In contrast, direct mid-air interaction appears better suited for brief, spatially direct actions that do not require entering a dedicated interaction mode. Future multi-window interfaces may therefore benefit from combining both modalities, allowing users to naturally choose the interaction technique that best matches the requirements of the current task, especially during complex and compound window-based workflows.

5.4 Posture Flexibility Through Hand Remapping

Input remapping allows users to interact with virtual windows from physical locations that are not perfectly aligned with the displayed content. Participants appreciated the ability to lower or rest their arms while continuing to interact with elevated virtual windows. However, two participants noted that they sometimes forgot to take advantage of this flexibility when concentrating on the task. These observations complement prior input-remapping techniques that decouple physical hand position from the location of virtual content [1, 8, 13, 57]. While prior work has demonstrated the ergonomic potential of redirected interaction, our findings highlight an additional practical consideration: users must adapt their interaction behavior to take advantage of the posture flexibility, and this flexibility remains bounded by the tracking volume of current hand-tracking systems.

These observations suggest that while input remapping provides greater posture flexibility, it also changes users' interaction strategies and requires adaptation. Future systems may therefore benefit from posture-aware guidance, tracking-aware interaction boundaries, or adaptive remapping techniques that encourage comfortable interaction without requiring users to consciously manage their physical posture. Such support may become increasingly important as hand tracking continues to improve and spatial computing systems support longer periods of interaction.

5.5 Limitations and Future Work

While participants generally appreciated using non-dominant hand gestures to set the interaction frame and context, the flexibility of unconstrained hand motion sometimes introduced excessive degrees of freedom during window manipulation. In particular, rotating application windows in the Over-Snap mode was occasionally perceived as difficult to control. Future iterations could incorporate interaction constraints, such as axis locking or stabilization techniques, to improve controllability while preserving the fluidity of direct manipulation. In addition, our current gesture recognition relies on fixed thresholds for joint flexion and inter-finger angles across all participants. Although these thresholds were selected through pilot testing, adaptive or personalized calibration could better accommodate variations in hand size, finger flexibility, and natural resting poses.

HandPad employs a lightweight touch detection method based on hand tracking, enabling real-time touch interaction on the hand surface without requiring external cameras. Although participants did not report noticeable inaccuracies during the study, more advanced vision-based approaches may further improve robustness under challenging tracking conditions. For example, recent work has demonstrated the potential of vision transformers for on-skin touch detection [42], suggesting a promising direction for future hand-surface interaction techniques.

While HandPad enables users to interact with a lower physical posture through input remapping, users require time to adapt to this interaction strategy and may not always take advantage of it when concentrating on demanding tasks. Future systems could provide subtle visual cues or adaptive guidance that encourage comfortable interaction postures without disrupting the workflow. Furthermore, the effectiveness of remapping is ultimately constrained by the tracking volume of current optical hand-tracking systems, which may limit how far users can lower their hands before tracking quality degrades. As hand-tracking technology continues to improve, these constraints are expected to become less restrictive.

While all participants in our study had prior experience with VR, only two reported frequent VR use, and the remaining participants reported only occasional use. Therefore, participants' reflections on HandPad relative to controller-based or mid-air interaction should be interpreted as subjective impressions based on their prior experiences rather than systematic comparisons. Future studies could recruit participants with more extensive experience using existing VR interaction techniques and directly compare HandPad with these approaches.

Finally, our evaluation was designed as an exploratory study of HandPad as an integrated interaction framework for realistic multi-window workflows in VR. While the findings provide insight into participants' experiences with the complete workflow, the study relied primarily on subjective ratings and qualitative feedback and was not intended to establish comparative advantages over existing interaction techniques. Future work should complement this exploratory evaluation with controlled studies that investigate individual components of the framework, such as palm interaction, mode transitions, input remapping, and window management, while incorporating objective measures of task performance, workload, and physical exertion to systematically evaluate HandPad's efficiency, accuracy, and ergonomic effects.

6 Conclusion

We presented HandPad, an asymmetric bimanual framework that organizes bare-hand interaction with multiple virtual windows across three levels: cross-window navigation, within-window content manipulation, and application-window management. The non-dominant hand establishes the reference frame, sets the interaction context, provides a hand-based touch surface, and anchors hand remapping, while the dominant hand performs on-hand touch and complementary mid-air actions relative to that context. Our exploratory study investigated participants' experiences using these techniques as an integrated multi-window workflow. Participants reported positive perceptions of ownership, agency, and tactile correspondence, while perceived ease of use varied across individual interaction techniques. Their feedback highlighted the stability provided by the hand surface and the ability to perform fluid multi-touch and cross-window interactions. Our findings further emphasize the potential of combining hand-surface interaction with complementary mid-air gestures to support interactions at different levels, where the suitability of each modality depends on the duration, task requirements, and spatial scope of the operation.

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A Touch-State-Based Manipulation Gesture Recognition

HandPad recognizes DH touch gestures on the NDH by combining the temporal state of each contact point with the identity of the contacting fingers. Once contact between a DH manipulation fingertip and the NDH palm is detected, each contact is classified into one of three states: touched, dwelling, or moving, following prior touchpad-based approaches [33].

Specifically, a touch enters the dwelling state if the contact point remains within the same location for longer than 450 ms. A touch or dwelling enters the moving state if the displacement of the contact point exceeds 10 mm. The touch state of the manipulation fingertip at a given moment is represented as three consecutive numbers: the number of touch points (N_t), the number of dwelling points (N_d), and the number of moving points (N_m). A gesture is then recognized as a sequence of touch states $\{N_t N_d N_m\}$. Unlike traditional touchpads, which cannot easily distinguish individual fingers and may therefore produce gesture ambiguities [33], our system integrates touch-state sequences with the identities of specific DH fingers. This allows us to differentiate gestures performed on the NDH palm surface. As illustrated in Figure 8, the sequence $\{000, 100, 000\}$ represents a click gesture. The sequence $\{000, 100, 001\}$ represents a swipe gesture when performed by the thumb, or a drag gesture when performed by the index finger. The sequence $\{000, 200, 002\}$ represents a scale or rotate gesture when performed by the thumb and index finger, and a scroll gesture when performed by the index and middle fingers. In practice, scale and rotate interactions may produce additional state sequences if one finger remains in the dwelling state while the other moves.

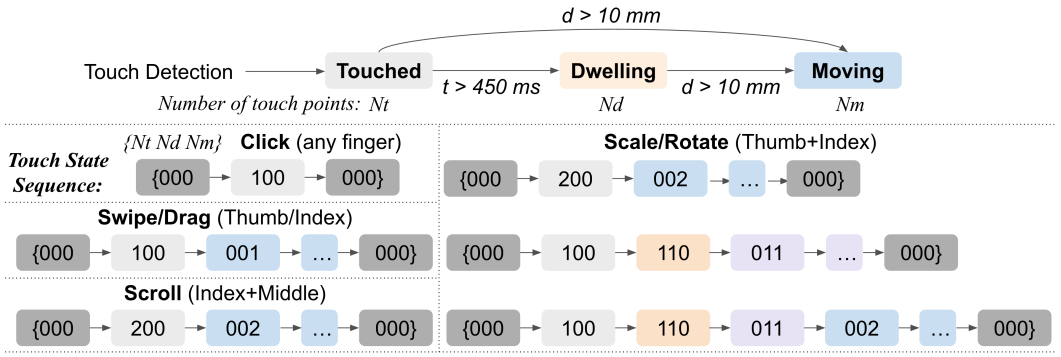


Fig. 8. Touch state representation and gesture sequences. Each touch point can be in one of three states: touched, dwelling, or moving. Each state is encoded by the number of points in these three phases. Example state sequences are shown for four gestures: click, scroll, swipe/drag, and scale/rotate.